

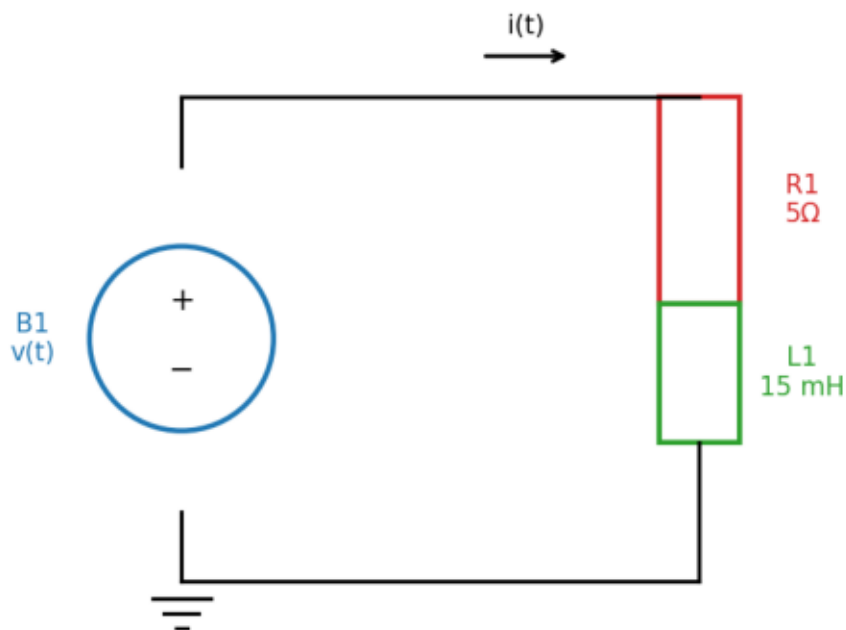
Exercise 12 — Non-sinusoidal Source, RL Load

A non-sinusoidal voltage source has a Fourier series of

$$v(t) = 10 + 20\cos(2\pi 60t - 25^\circ) + 30\cos(4\pi 60t + 20^\circ) \text{ V}$$

This voltage is connected to a load that is 5Ω resistor and 15 mH inductor in series. Determine the power absorbed by the load.

Exercise 12 schematic (as wired in exercise12_schematic.asc)
 $v(t) = 10 + 20\cos(2\pi 60t - 25^\circ) + 30\cos(4\pi 60t + 20^\circ) \text{ V}$



circuits\exercise12_schematic.asc — open directly in LTspice
(verified: simulated results match the hand solution below)

Solution (1/2) — Phasor current in each term

Step 1 — DC term ($\omega = 0$, inductor is a short circuit)

$$I_0 = V_0 / R = 10 / 5 = 2 \text{ A}$$

Step 2 — AC term at $\omega_1 = 2\pi(60) = 377 \text{ rad/s}$ (source: $20\angle -25^\circ \text{ V}$)

$$Z_1 = R + j\omega_1 L = 5 + j(377)(0.015) = 5 + j5.655 \Omega$$

$$|Z_1| = \sqrt{5^2 + 5.655^2} = 7.55 \Omega, \quad \angle Z_1 = \tan^{-1}(5.655/5) = 48.5^\circ$$

$$I_1 = V_1 / Z_1 = 20\angle -25^\circ / 7.55\angle 48.5^\circ = 2.65\angle -73.5^\circ \text{ A}$$

Step 3 — AC term at $\omega_2 = 4\pi(60) = 754 \text{ rad/s}$ (source: $30\angle 20^\circ \text{ V}$)

$$Z_2 = R + j\omega_2 L = 5 + j(754)(0.015) = 5 + j11.31 \Omega$$

$$|Z_2| = \sqrt{5^2 + 11.31^2} = 12.37 \Omega, \quad \angle Z_2 = \tan^{-1}(11.31/5) = 66.2^\circ$$

$$I_2 = V_2 / Z_2 = 30\angle 20^\circ / 12.37\angle 66.2^\circ = 2.43\angle -46.2^\circ \text{ A}$$

Result:

$$i(t) = 2 + 2.65\cos(2\pi 60t - 73.5^\circ) + 2.43\cos(4\pi 60t - 46.2^\circ) \text{ A}$$

LTspice check (Fourier of I(L1), transient sim):

$$\text{DC} = 1.9996 \text{ A}, \quad 60 \text{ Hz} = 2.649\angle 73.51^\circ, \quad 120 \text{ Hz} = 2.426\angle 46.15^\circ$$

(magnitudes match to 3 s.f.; angle sign is a reference-direction convention difference between the loop current and I(L1))

Solution (2/2) — Power absorbed by the load

Average power at each frequency: $P = (1/2)V \cdot I \cdot \cos(\theta_v - \theta_i)$

DC term:

$$P_0 = V_0 \cdot I_0 = (10)(2) = 20 \text{ W}$$

$\omega_1 = 2\pi(60)$:

$$\begin{aligned} P_1 &= (20)(2.65)/2 \cdot \cos(-25^\circ - (-73.5^\circ)) = 26.5 \cdot \cos(48.5^\circ) \\ &= 26.5 \times 0.6626 \approx 17.4 \text{ W} \end{aligned}$$

$\omega_2 = 4\pi(60)$:

$$\begin{aligned} P_2 &= (30)(2.43)/2 \cdot \cos(20^\circ - (-46.2^\circ)) = 36.45 \cdot \cos(66.2^\circ) \\ &= 36.45 \times 0.4014 \approx 14.8 \text{ W} \end{aligned}$$

Total power absorbed by the load:

$$P = P_0 + P_1 + P_2 = 20 + 17.4 + 14.8 = 52.2 \text{ W}$$

Alternative check (rms current method, average inductor power = 0):

$$P = I_{\text{rms}}^2 \cdot R = [2^2 + (2.65/\sqrt{2})^2 + (2.43/\sqrt{2})^2] \cdot (5) = 52.2 \text{ W}$$

LTspice check (numeric time-average of $v(t) \cdot i(t)$ over one exact 60 Hz period, after the RL turn-on transient settles):

$$P_{\text{avg}} (\text{simulated}) = 52.27 \text{ W} \quad \text{— matches the hand solution.}$$

Simulated waveforms (LTspice, steady state)

Exercise 12 - Non-sinusoidal source, $R=5\Omega$ + $L=15\text{mH}$ series load

